The documentation and process conversion measures necessary to comply with this revision shall be completed by 17 January 2013.

INCH-POUND

MIL-PRF-19500/680A
17 October 2012
MIL-PRF-19500/680
7 September 2000

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, DUAL SCHOTTKY CENTER TAP
POWER RECTIFIER, SURFACE MOUNTED,
TYPE 1N6842U3, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, dual Schottky power rectifier in a surface mount package. Four levels of product assurance are provided for each device types as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (U3).

1.3 Maximum ratings.

| Type | $V_{RWM}$ | $I_O$ (1) $T_C$= +100°C | $I_{FSM}$ $T_C$ = +25°C $T_p$ = 8.3 ms | $C_J$ at 5 V | $T_{STG}$ | $T_J$ |
|---|---|---|---|---|---|---|
| | V | A dc | A (pk) | pF | °C | °C |
| 1N6842U3 | 60 | 10 | 200 | 400 | -65 to +150 | -65 to +150 |

(1) Derate linearly at 200 mA/°C from $T_J = T_C$ =+100°C to +150°C.

1.4 Primary electrical characteristics. $R_{\theta JC}$ = 2.8ºC/W maximum on each leg.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.



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## 2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or https://assist.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3 REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (SMD.5) herein. The US government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

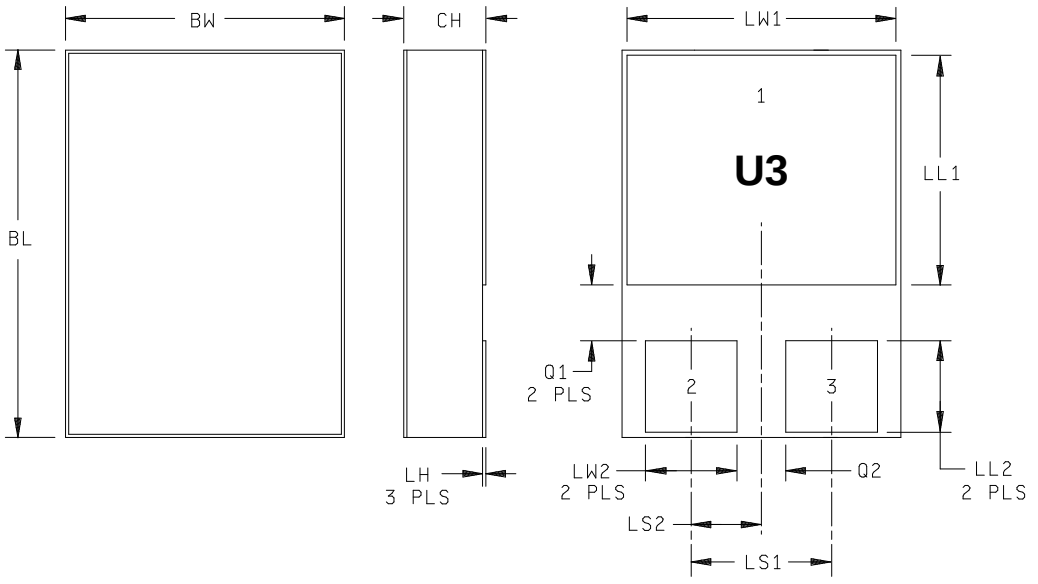
* 3.4.1 Polarity and lead finish. Polarity and terminal configuration shall be in accordance with figure 1 herein. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

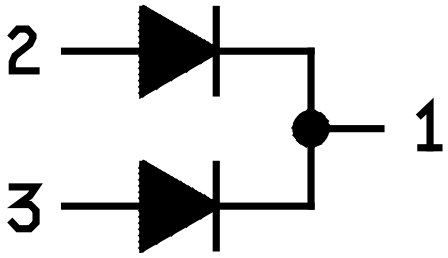
3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking may be omitted from the body, but shall be retained on the initial container.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.



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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .395 | .405 | 10.03 | 10.29 |
| BW | .291 | .301 | 7.39 | 7.65 |
| CH | .108 | .124 | 2.74 | 3.15 |
| LH | .010 | .020 | 0.25 | 0.51 |
| LL1 | .220 | .230 | 5.59 | 5.84 |
| LL2 | .115 | .125 | 2.92 | 3.18 |
| LS1 | .150 BSC | | 3.81 BSC | |
| LS2 | .075 BSC | | 1.91 BSC | |
| LW1 | .281 | .291 | 7.14 | 7.39 |
| LW2 | .090 | .100 | 2.29 | 2.54 |
| Q1 | .030 | | 0.76 | |
| Q2 | .030 | | 0.76 | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.
4. Terminal 1 is common cathode.
5. Terminal 2 is anode 1.
6. Terminal 3 is anode 2.

* FIGURE 1. Physical dimensions and configuration (SMD.5).



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3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I and II herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4 VERIFICATION

* 4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

* c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.1 Group E inspection. Group E inspection shall be in accordance with herein.

* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) 3c | Method 3101 of MIL-STD-750 (see 4.3.2) | Method 3101 of MIL-STD-750 (see 4.3.2) |
| 9 and 10 | Not applicable | Not applicable |
| 11 | $V_{F2}$ and $I_{R1}$ | $V_{F2}$ and $I_{R1}$ |
| 12 | Condition A, see 4.3.1, t = 240 hours | Condition A, see 4.3.1, t = 48 hours |
| 13 | Subgroup 2 and 3 of table I herein; $\Delta V_{F2}$ = ±50 mV, $\Delta I_{R1}$ = 100 percent of initial value or 25 µA, whichever is greater. | Subgroup 2 of table I herein; $\Delta V_{F2}$ = ±50 mV, $\Delta I_{R1}$ = 100 percent of initial value or 25 µA, whichever is greater. |

* (1) Shall be performed anytime after screen 3a. JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Burn-in conditions are as follows: Method 1038 of MIL-STD-750, test condition A. $T_J$ = +125°C minimum; $V_R$ = 80 percent of rated $V_{RWM}$ dc.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, (and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 µs max. See table III, group E, subgroup 4 herein.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with table II herein.

* 4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 4066 | $I_{FSM}$ = 200 A for each leg; one surge of 8.3 ms superimposed on $I_O$. Condition A, $T_A$ = +25°C; $V_R$ = 0; $I_O$ = 10 A continuous half-wave. |
| B4 | 1037 | Condition for intermittent operation life for each diode is as follows: $I_F$ or $I_O$ = 1 A (minimum); $\Delta T_J$ = +85°C, +15°C, -5°C for 2,000 cycles minimum. |
| B5 | 1038 | Condition for steady-state operation life (accelerated) is as follows: Conditions B, $T_j$ = +150°C, $T_A$ +100°C, 240 hrs. |
| * B6 | 3101 | Limit for thermal resistance is $R_{\theta JC}$ = 2.80ºC/W for each side. |

* 4.4.2.2 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 4066 | $I_{FSM}$ = 200 A for each leg; one surge of 8.3 ms superimposed on $I_O$. Condition A, $T_A$ = +25°C; $V_R$ = 0; $I_O$ = 10 A continuous half-wave. |
| B3 | 1037 | Condition for intermittent operation life for each diode is as follows: $I_F$ or $I_O$ = 1 A (minimum); $\Delta T_J$ = 85°C, +15°C, -5°C for 2,000 cycles minimum. |
| B5 | | Not applicable. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with table II herein.

| Subgroup | Method | Conditions |
|---|---|---|
| * C5 | 3101 | Limit for thermal resistance is $R_{\theta JC}$ = 2.80ºC/W for each side. |
| C6 | 1037 | $I_F$ or $I_O$ = 1A (minimum). |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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* TABLE I. Group A inspection. 1/ 2/

| Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| * Thermal impedance | 3101 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Forward voltage | 4011 | | | | | |
| | | $I_{FM}$ = 3 A (dc) pulsed (see 4.5.1) | $V_{F1}$ | | 0.62 | V dc |
| | | $I_{FM}$ = 10 A (dc) pulsed (see 4.5.1) | $V_{F2}$ | | 0.78 | V dc |
| | | $I_{FM}$ = 15 A (dc) pulsed (see 4.5.1) | $V_{F3}$ | | 0.90 | V dc |
| Reverse current leakage 1N6842U3 | 4016 | DC method, pulsed (see 4.5.1.) $V_R$ = 60 V dc | $I_{R1}$ | | 50 | uA dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +100°C | | | | |
| Reverse current leakage 1N6842U3 | 4016 | DC method, pulsed (see 4.5.1) $V_R$ = 60 V dc | $I_{R2}$ | | 10 | mA dc |
| Forward voltage | 4011 | $I_F$ = 10 A pulsed (see 4.5.1) | $V_{F4}$ | | 0.70 | V dc |
| | | $I_F$ = 15 A pulsed (see 4.5.1) | $V_{F5}$ | | 0.80 | V dc |
| Forward voltage | 4011 | $T_A$ = -55°C | | | | |
| | | Pulsed (see 4.5.1) $I_F$ = 10 A (dc) | $V_{F6}$ | | 0.85 | V dc |
| Subgroup 4 | | | | | | |
| Junction capacitance | 4001 | $V_R$ = 5 V dc; f = 1 MHz, $V_{SIG}$ = 50 mV (p-p) (max) | $C_J$ | | 400 | pF |
| Subgroup 5 | | | | | | |
| Insulation resistance | 1016 | $V_R$ = 600 V dc; from lid to bottom case. All terminals shorted. | $I_{RES}$ | | 10 | µA |
| Subgroup 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ All measurements are for each side.



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* TABLE II. Group B, C, and E, electrical end-point measurement 1/ 2/ 3/

| | Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|
| | | | Method | Conditions | | Min | Max | |
| | 1. | Forward voltage | 4011 | $I_F$ = 10 A (pk) pulsed (see 4.5.1) | $\Delta V_{F2}$ | | ±50 | mV |
| | 2. | Reverse current leakage | 4016 | Pulsed (see 4.5.1) DC method $V_R = V_{RWM}$ | $\Delta I_{R1}$ | | 25 µA or 100 percent of initial reading, whichever is greater. | |
| * | 3. | Thermal impedance | 3101 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |

1/ The electrical measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table II herein, steps 1 and 2.
   b. Subgroup 4, see table II herein, steps 1, 2, and 3.
   c. Subgroup5, see table II herein, steps 1, 2, and 3.

* 2/ The electrical measurements for table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, steps 1 and 2.
   b. Subgroup 3, see table II herein, steps 1, 2, and 3.

3/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, steps 1 and 2 for all levels.
   b. Subgroup 6, see table II herein, steps 1, 2 and 3 for all levels.



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* TABLE III. Group E inspection (all quality levels) for qualification only.

| Inspection | MIL-STD-750 | | Qualification inspection |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices, c = 0 |
| Thermal shock (temperature cycling) | 1051 | 500 cycles. Condition G. | |
| Hermetic seal | 1071 | | |
| Electrical measurements | | See table I, subgroup 2, and table II, steps 1 and 2. | |
| Subgroup 2 | | | 45 devices, c = 0 |
| Steady-state blocking life | 1038 | t = 1,000 hours, $T_J$ = +125°C; $V_R$ = 80 percent of rated $V_{RWM.}$ | |
| Electrical measurements | | See table I, subgroup 2, and table II, steps 1 and 2. | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | Sample size N/A |
| * Thermal impedance curves | 3101 | See MIL-PRF-19500. | |
| Subgroup 5 | | | 5 devices, c = 0 |
| Surge | 4066 | Condition A, $T_A$ = +25°C; $I_{FSM}$ = 200 A for each leg, one surge of 8.3 ms superimposed on $I_O$. $V_R$ = 0; $I_O$ = 10 A half sine wave, continuous. | |
| Electrical measurements | | See table I, subgroup 2, and table II, steps 1 and 2. | |



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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil .

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activity:
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